

Data Catalog for Gold Layer

The Gold Layer is the business-level data representation of the Vidhya SQL Data Warehouse. It is designed to support analytical, reporting, and business intelligence requirements.

The Gold Layer contains dimension tables and fact tables that organize customer, product, and sales information for easy analysis.

1. gold.dim_customers

Purpose: Stores customer information enriched with demographic and geographic attributes.

Column Name	Data Type	Description
customer_key	INT	Surrogate key that uniquely identifies each customer record in the dimension table.
customer_id	INT	Unique identifier assigned to the customer.
customer_number	NVARCHAR(50)	Alphanumeric customer identifier used for tracking and reference.
first_name	NVARCHAR(50)	Customer's first name.
last_name	NVARCHAR(50)	Customer's last name or family name.
country	NVARCHAR(50)	Country associated with the customer.
marital_status	NVARCHAR(50)	Marital status of the customer.
gender	NVARCHAR(50)	Gender of the customer.
birthdate	DATE	Customer's date of birth.
create_date	DATE	Date on which the customer record was created.

2. gold.dim_products

Purpose: Stores product information and attributes required for product analysis.

Column Name	Data Type	Description
product_key	INT	Surrogate key that uniquely identifies each product record.
product_id	INT	Unique identifier assigned to the product.
product_number	NVARCHAR(50)	Alphanumeric code used to identify and track the product.
product_name	NVARCHAR(50)	Name or description of the product.
category_id	NVARCHAR(50)	Identifier representing the product category.
category	NVARCHAR(50)	High-level classification of the product.
subcategory	NVARCHAR(50)	Detailed classification of the product within its category.
maintenance_required	NVARCHAR(50)	Indicates whether maintenance is required for the product.
cost	INT	Base cost of the product.
product_line	NVARCHAR(50)	Product line or series to which the product belongs.
start_date	DATE	Date from which the product became available.

3. gold.fact_sales

Purpose: Stores transactional sales information used for sales analysis and reporting.

Column Name	Data Type	Description
order_number	NVARCHAR(50)	Unique identifier for each sales order.
product_key	INT	Surrogate key linking the sale to the product dimension.
customer_key	INT	Surrogate key linking the sale to the customer dimension.

Column Name	Data Type	Description
order_date	DATE	Date when the order was placed.
shipping_date	DATE	Date when the order was shipped.
due_date	DATE	Date when payment was due.
sales_amount	INT	Total sales value for the sales line.
quantity	INT	Number of units purchased.
price	INT	Price of one unit of the product.

Data Model Relationship

The Gold Layer follows a simple star-schema structure:

gold.dim_customers → Customer information

gold.dim_products → Product information

gold.fact_sales → Sales transactions

The "fact_sales" table connects to the dimension tables using:

- "fact_sales.customer_key" → "dim_customers.customer_key"
- "fact_sales.product_key" → "dim_products.product_key"

This structure makes the data easier to use for SQL analysis, Power BI dashboards, reporting, and business intelligence.

Project Owner

Project Name: Vidhya SQL Data Warehouse

Documentation: Data Catalog

Layer: Gold Layer